Amirkabir University of Technology

Department of Electrical Engineering

Report for Assignment Series 2

Adaptive Control (MRAC) Course

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Introduction

In this assignment, we study the design of a Model Reference Adaptive Controller (MRAC) for both linear and nonlinear systems. The goal is to make the real system's behavior match a given reference model.

Problem Statement

The linear system model is given as:

$$\dot{x_p} = 3x_p + 0.5u$$

The reference model is as follows:

$$\dot{x}_m = -2x_m + 2r$$

a) Design a direct adaptive controller.

b) Design an indirect adaptive controller.

c) Design a direct adaptive controller for the following nonlinear system:

$$\dot{x_p} = 3x + 2\sin^2(x_p) + x_p^3 + u$$

a) Direct Adaptive Control

For designing a direct adaptive controller, define:

$$e = x_p - x_m$$

where x_p is the real system output and x_m is the reference model output.

The control input is chosen as:

$$u = Kr + \theta x_p$$

where K and θ are adaptive parameters.

The ideal values of these coefficients are defined as:

$$K^* = \frac{k_m}{k_p} \qquad \theta^* = \frac{a_m - a_p}{k_p}$$

where a_m, k_m are reference model parameters and a_p, k_p are real system parameters.

Using a suitable Lyapunov function and Lyapunov-based adaptation, the update laws are:

$$\dot{\psi} = \dot{K} = -\beta_1 \operatorname{sgn}(k_p)\, e\, r$$
$$\dot{\phi} = \dot{\theta} = -\beta_2 \operatorname{sgn}(k_p)\, e\, x_p$$

where β_1 and β_2 are positive adaptation gains. In the final simulation, these are set to 4.5 and 5, respectively.

b) Indirect Adaptive Control

In the indirect method, system parameters are first estimated and then used in the control law.

Define the error signal as:

$$e = \hat{x}_p - x_p$$

where $\hat{x}_p$ is the output of the adaptive estimator.

The control input remains:

$$u = Kr + \theta x_p$$

However, K and θ are computed indirectly from $\hat{a}_p$ and $\hat{k}_p$.

The adaptive model of the system is defined as:

$$\dot{\hat{x}}_p = a_m \hat{x}_p + (\hat{a}_p - a_p) x_p + \hat{k}_p u$$

Update laws for parameters:

For estimating a_p:

$$\dot{\hat{a}}_p = -\beta_2\, e\, x_p$$

where β_2 is a positive constant.

To ensure $\bar{k_p} > k_{p_{min}}$ (e.g., $k_{p_{min}} = 0.1$) to avoid control instability, a conditional update law is used:

$$\dot{\hat{k}}_p = f(e, u, \hat{k}_p, \bar{k_p})$$

Function f is defined as:

- If $\hat{k}_p > k_{p_{min}}$:
$$\dot{\hat{k}}_p = -\beta_1 \, e \, u$$
- Else:
  - If $\beta_1 \, e \, u > 0$:
  $$\dot{\hat{k}}_p = -\beta_1 \, e \, u$$
  - Else:
  $$\dot{\hat{k}}_p = 0$$

The initial value $\hat{k}_p(0)$ should be greater than $k_{p_{min}}$ (e.g., 0.2) to ensure stable control.

The gains β_1 and β_2 are set to 1 and 8, respectively, in the final simulation.

c) Nonlinear System Control

Now, we design a direct adaptive controller for the nonlinear system:

$$\dot{x_p} = 3x_p + 2\sin^2(x_p) + x_p^3 + u$$

Define the error:

$$e = x_p - x_m$$

Control input:

$$u = Kr + \theta_1 x_p + \theta_2 f(x_p)$$

where $f(x_p) = \sin^2(x_p) + x_p^3$, and K, θ_1, θ_2 are adaptive parameters.

Ideal values:

$$K^* = \frac{k_m}{k_p}, \quad \theta_1^* = \frac{a_m - a_p}{k_p}, \quad \theta_2^* = \frac{-\alpha}{k_p}$$

Update laws based on Lyapunov theory:

$$\dot{K} = -\beta_1 \operatorname{sgn}(k_p)\, e\, r$$

$$\dot{\theta}_1 = -\beta_2 \operatorname{sgn}(k_p)\, e\, x_p$$

$$\dot{\theta}_2 = -\beta_3 \operatorname{sgn}(k_p)\, e\, f(x_p)$$

Where $\beta_1 = 1$, $\beta_2 = 3$, and $\beta_3 = 5.5$. These values are tuned for sinusoidal inputs with frequencies 1, 2, and 10; they should be adjusted for other inputs.

Simulation: Direct Adaptive Control for Linear System

This section presents the simulation results for direct adaptive control applied to the linear system. The Simulink model and results are as follows:

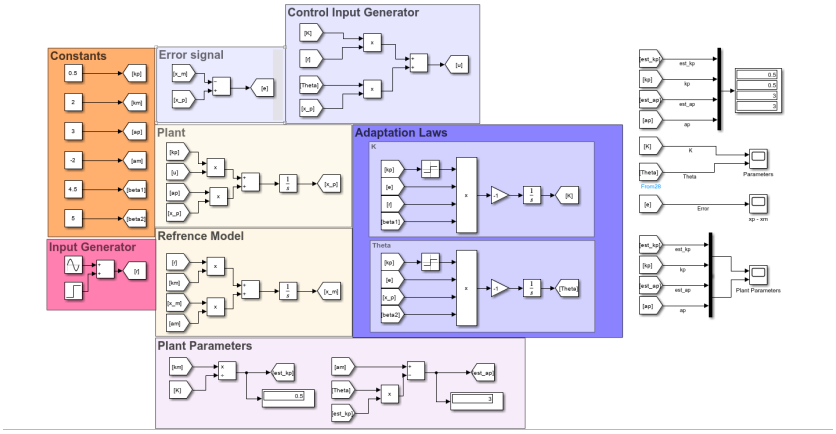


Figure 1: Overview of the Simulink model for direct adaptive control

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System Behavior Under Different Inputs

This section investigates how the reference input signal $r(t)$ affects adaptive system performance. Two types of inputs are used:

- **Sinusoidal Input:** $r(t) = \sin(t)$. In direct control, the system converges well to the reference model, and the error $e = x_p - x_m$ tends to zero. In indirect control, due to conditional update rules (if-else) for $\hat{k}_p$, perfect convergence is not achieved, and a small steady-state error remains.

- **Step Input:** A positive step function. The system does not fully converge to the reference model, and a nonzero steady-state error remains. This is due to the sudden change and discontinuity in the derivative of the step signal, which challenges the system's ability to track the reference model.

Overall, the results show that adaptive controllers perform better with smooth, continuous inputs (like sinusoidal signals) compared to discrete or non-differentiable ones (like step inputs).

Conclusion

In this assignment, the design and implementation of direct and indirect Model Reference Adaptive Controllers (MRAC) were studied for linear and nonlinear systems. The simulation results demonstrate that adaptive controllers can compensate for parameter uncertainties and changes in the system and provide acceptable performance.

In the direct method, proper adaptation laws ensured good convergence to the reference model without explicit parameter estimation. In the indirect method, although protective mechanisms (such as if-else conditions) were used to prevent unstable parameter updates, full convergence was not achieved, leaving a small steady-state error.

Additionally, testing the controllers under different reference inputs showed that the adaptive system performs better with smooth and differentiable inputs.